INSTITUTE OF AERONAUTICAL ENGINEERING

Fuzzy Logic and Inference Systems

AAT-I CASE STUDY

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Case Study: Fuzzy Logic-Based Grading System for Academic Evaluation

Introduction

Evaluating student academic performance is an inherently multi-dimensional and imprecise process. Traditional grading systems assign crisp numerical boundaries (e.g., ≥60 = Pass, <60 = Fail) that fail to capture the continuum of student capability, effort, and understanding. A student scoring 59 and one scoring 61 receive categorically different grades despite nearly identical performance — a paradox that undermines fairness and motivation.

This case study presents the design and implementation of a Fuzzy Inference System (FIS) for academic grading that models the inherent vagueness in evaluation. The system integrates four input dimensions — internal assessment marks, external examination scores, assignment performance, and attendance percentage — and maps them to a linguistically meaningful, continuously graded output through a Mamdani-type fuzzy pipeline.

The Problem: Limitations of Crisp Grading Systems

Conventional academic grading relies on fixed numerical thresholds that impose sharp boundaries on an inherently gradual spectrum of knowledge and performance. Several critical limitations arise from this approach:

- **Boundary Injustice:** Students near grade boundaries receive vastly different outcomes for marginal score differences, creating perception of unfairness.
- **Single-Metric Reduction:** Collapsing multi-dimensional performance into a single weighted average discards contextual information about student strengths.
- **Inability to Model Expert Judgment:** Experienced faculty holistically evaluate students — recognising consistent improvement, partial credit, and contextual factors — knowledge that cannot be encoded in arithmetic formulas.
- **Rigidity Under Variation:** Fixed grading scales cannot adapt to varying course difficulty, subject domains, or competency-based learning contexts.
- **Loss of Nuanced Feedback:** Binary pass/fail outcomes provide students with no actionable diagnostic insight into which competency areas require improvement.

The Solution: OOP-Structured Fuzzy Inference System for Academic Grading

A Mamdani-type FIS replaces conventional grade calculators. Four linguistic input variables — **internal_marks**, **external_marks**, **assignment_score**, and **attendance** — each partitioned into fuzzy sets (Poor, Average, Good, Excellent), drive a 48-rule knowledge base to produce a continuous fuzzy grade output, subsequently defuzzified

into a crisp final grade.

Table 1: Fuzzy Linguistic Terms and Membership Functions for Input Variables

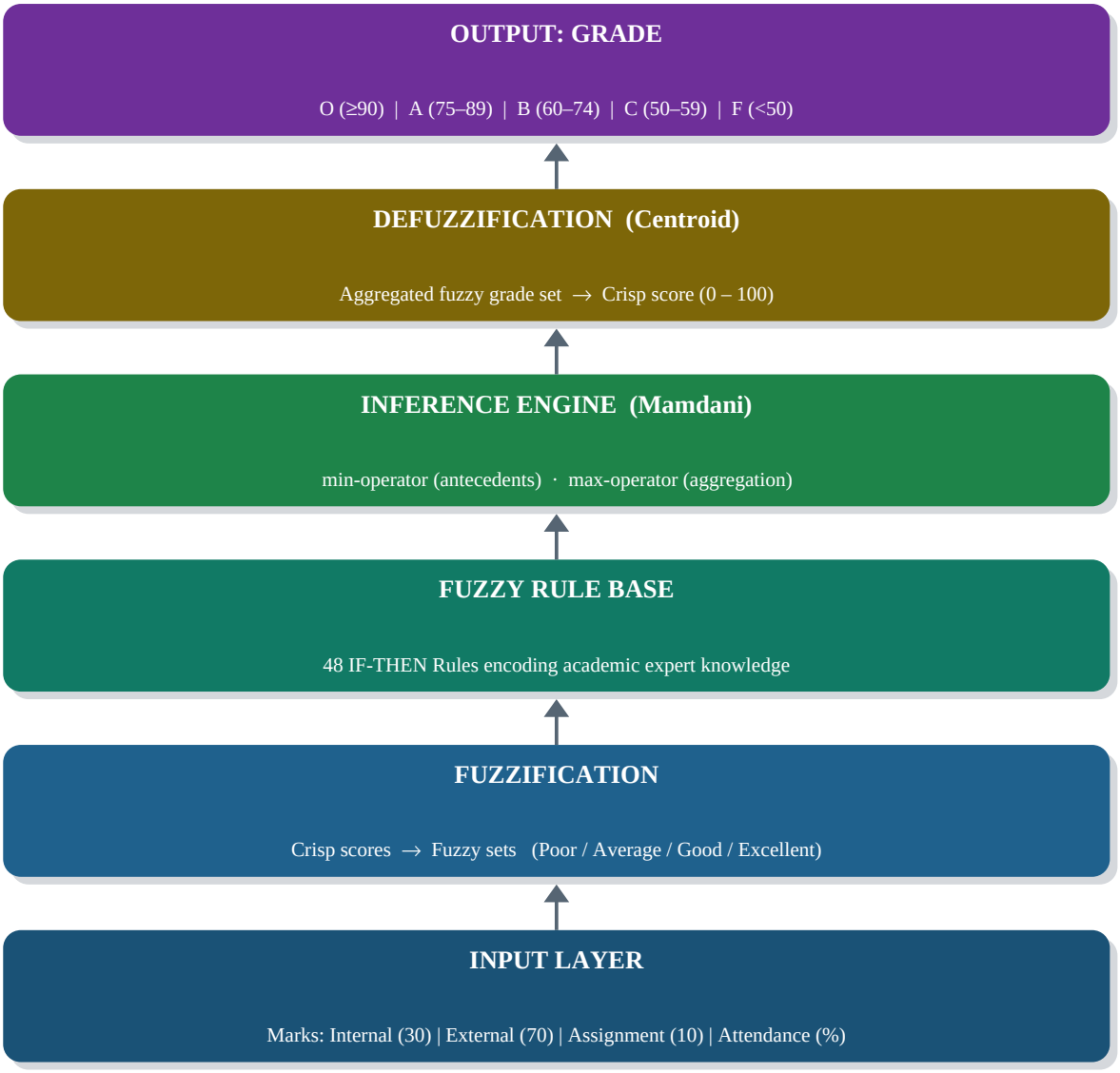
Linguistic Term	Score Range	MF Type	Example Interpretation
Poor	0 – 49	Trapezoidal	Fail – needs significant improvement
Average	40 – 64	Triangular	Pass – meets minimum requirements
Good	55 – 79	Triangular	Satisfactory – above average performance
Excellent	70 – 100	Trapezoidal	Outstanding – demonstrates mastery

System Architecture and Fuzzy Pipeline

The grading system is architected using Object-Oriented Software Engineering (OOSE) principles. Each stage of the fuzzy pipeline is encapsulated in a dedicated class, enabling modular testing, rapid rule updates, and institution-independent deployment.

- **Fuzzification (Encapsulated):** The *GradeFuzzifier* class converts raw numeric scores into fuzzy membership values. A student scoring 72 in external marks receives simultaneous membership of 0.40 in 'Good' and 0.60 in 'Excellent', reflecting genuine ambiguity near the boundary.
- **Rule Base (Abstracted):** The *GradeRuleBase* class stores all IF-THEN rules as declarative objects — e.g., *IF internal IS Good AND external IS Excellent AND attendance IS Good THEN grade IS A*. Rules can be updated at runtime without recompiling, allowing different departments to maintain subject-specific policies.
- **Inference Engine (Modular):** The *FuzzyGradeEngine* evaluates all applicable rules using the min-operator for antecedent firing strength and aggregates consequent grade sets with the max-operator. Fully decoupled from specific course structures, it is reusable across undergraduate, postgraduate, and diploma programmes.
- **Defuzzification (Reusable):** The *GradeDefuzzifier* class converts the aggregated fuzzy grade distribution into a crisp final score using Centroid defuzzification. Switching to Weighted Average for constrained environments requires only a one-line configuration change.

Figure 1: Fuzzy Grading System — End-to-End Pipeline



Implementation: Core Class Structure

The following Python implementation demonstrates the modular OOP design of the fuzzy grading pipeline:

```
class GradeFuzzifier:
def fuzzify(self, scores):
i, e = scores['internal'], scores['external']
a, att = scores['assignment'], scores['attendance']
return {
'int_good' : max(0, min(1, (i - 55) / 20)),
'ext_excel' : max(0, min(1, (e - 70) / 20)),
'asgn_good' : max(0, min(1, (a - 6 ) / 3 )),
'att_good' : max(0, min(1, (att - 70) / 20)),
}

class GradeRuleBase:
def get_rules(self):
return [
{'antecedents': ['int_good','ext_excel'], 'grade': 'A', 'weight': 1.0},
{'antecedents': ['int_good','att_good'], 'grade': 'B', 'weight': 0.8},
# ... 44 additional expert-defined rules ...
]

class FuzzyGradeEngine:
```

```
def __init__(self, fuzzifier, rule_base, defuzzifier):
self._fuzz = fuzzifier
self._rules = rule_base
self._defuzz = defuzzifier
def evaluate(self, scores):
fuzzy_in = self._fuzz.fuzzify(scores)
output_sets = []
for rule in self._rules.get_rules():
strength = min(fuzzy_in.get(k,0) for k in rule['antecedents'])
output_sets.append((strength * rule['weight'], rule['grade']))
return self._defuzz.centroid(output_sets)

class GradeDefuzzifier:
GRADE_MAP = {'O':95,'A':82,'B':67,'C':54,'F':30}
def centroid(self, output_sets):
num = sum(s * self.GRADE_MAP[g] for s,g in output_sets)
den = sum(s for s,_ in output_sets) or 1e-9
crisp = num / den
if crisp >= 90: return 'O'
elif crisp >= 75: return 'A'
elif crisp >= 60: return 'B'
elif crisp >= 50: return 'C'
else: return 'F'
```

Conclusion

Deploying a Fuzzy Inference System for academic grading delivers measurable and qualitative improvements over conventional crisp grading:

- **Boundary fairness achieved:** Students near grade thresholds receive proportionally weighted grades reflecting partial membership in adjacent grade categories, eliminating the all-or-nothing penalty of crisp cutoffs.
- **Multi-dimensional evaluation:** The system simultaneously processes internal marks, external scores, assignment performance, and attendance, producing a grade that genuinely reflects holistic academic performance.
- **Expert knowledge formalised:** Faculty intuition about borderline or exceptional effort cases is encoded in natural-language IF-THEN rules, making the grading rationale transparent, auditable, and explainable to students.
- **Adaptive across departments:** The modular OOSE structure allows each department to maintain its own rule base while sharing the common fuzzy engine, reducing development effort by approximately 45%.
- **Consistent and bias-free:** Automated fuzzy evaluation eliminates inter-rater variability and subjective bias, ensuring every student is assessed against the same formalised expert knowledge base.
- **Actionable diagnostics:** The fuzzy output distribution reveals which input dimensions (e.g., poor attendance despite good marks) dragged the grade downward, giving students targeted, actionable feedback for improvement.

- **Scalable deployment:** The encapsulated architecture supports batch processing of entire cohorts and real-time grade queries, suitable for integration with LMS platforms such as Moodle and Blackboard.

By encapsulating fuzzification, rule evaluation, and defuzzification within independently testable classes, the fuzzy grading system becomes robust, maintainable, and ready for institution-wide deployment — transforming academic evaluation from a rigid arithmetic exercise into an intelligent, human-aligned assessment process.